

GLOBAL SUPERSTORE PROJRCT REPORT

Project Overview:

This project analyzes the Global Superstore dataset using SQL and Power BI. The objective is to identify sales trends, customer purchasing behaviour, product performance, profitability, and regional performance to support business decision-making.

Problem Statement:

- Identify which countries and regions generate the highest sales.
- Determine which product categories contribute the most profit.
- Analyze whether high-frequency customers generate more revenue and profit.
- Find out which customer segments are the most profitable.
- Examine how customers are distributed across different countries.
- Identify the top-performing products based on sales and profit.
- Calculate the average delivery time across products.

Dataset Information:

Dataset: Global Superstore

- row_id
- order_id
- order_date
- ship_date
- ship_mode
- customer_id
- customer_name
- segment
- city
- state
- country
- Global market
- Region
- product_id
- category

- sub_category
- product_name
- sales
- quantity
- discount
- profit
- shipping_cost
- order_priority

DATA CLEANING & DATA PREPARATION:

Checked the dataset structure and verified all columns.

- Reviewed data quality and identified missing values.
- Found missing values in Postal Code.
- Removed Postal Code because it was not required for analysis there too many missing values.
- Cleaned customer names by removing extra spaces.
- Cleaned city names for consistency.
- Standardized country names to avoid duplicates.
- Cleaned product names for accurate reporting.
- Removed unnecessary data to improve performance.
- Prepared clean data for analysis and dashboard creation.
- Improved overall data accuracy and consistency.
- Converted string data in date format(STR_TO_DATE(order_date,'%d-%m-%Y') AS order_date,STR_TO_DATE(ship_date,'%d-%m-%Y') AS ship_date) in new table.

ETL & FACT TABLE CREATION:

- Converted raw transactional data into a structured fact table for analysis.
- Transformed order and shipping dates into standard date format.

- Combined customer, product, order, and sales data into a centralized table.
- Prepared data for KPI reporting and business analysis.
- Enabled time-based analysis and dashboard creation.
- Created a warehouse-ready structure for Power BI reporting.

We create dimensional tables for better analysis as below:

Dim_Customer:

- Customer_ID
- Customer_Name
- Segment
- City
- State
- Country
- Market
- region

Dim_Order:

- order_id
- order_date
- ship_date
- ship_mode
- order_priority

Dim_Product:

- product_id
- category
- sub_category
- product_name

- quantity

fact_sales:

This table create for schemas

- row_id
- order_id
- customer_id
- product_id
- sales
- quantity
- discount
- profit
- shipping_cost

DATA MODELING (STAR SCHEMA):

- Designed a Star Schema model for efficient reporting and analytics.
- Created separate dimension tables for Customer, Product, and Order data.
- Built a centralized Fact Sales table to store business transaction metrics.
- Reduced data redundancy by separating descriptive and transactional data.
- Improved query performance and dashboard responsiveness.
- Prepared an optimized structure for KPI reporting and Power BI analysis.

SQL Analysis:

1. Sales Analysis

Monthly Sales

- Monthly sales performance was analyzed using Group by Month(order_date).

- Sales trends were identified month-wise to determine peak and low-performing months.
- Monthly analysis helps in understanding seasonal demand patterns and sales fluctuations.

Yearly Sales

- Annual sales performance was calculated using Group by Year(order_date).
- Year-over-year growth and revenue changes were evaluated.
- This analysis supports long-term business performance measurement.

2. Product Analysis

Top Revenue Products

- Products generating the highest revenue were identified.
- Analysis was performed using Order By sum(sales) Desc.
- Top 10 products were selected using LIMIT 10.
- These products contributed significantly to overall business revenue.

Loss-Making Products

- Products with negative profitability were identified.
- Analysis used Having sum(Profit)<0.
- These products require review for pricing, discounts, or cost optimization.

3. Customer Analysis

Top Customers by Profit

- Customers contributing the highest profit were identified.
- Customer profitability helped determine valuable customer segments.

Repeat Customer Ranking

- Customer purchase frequency and ranking were analyzed.
- Window functions were applied:

- RANK()-Assigns rankings with skipped positions in case of ties.
 - DENSE_RANK()-Assigns rankings without skipping positions.
 - ROW_NUMBER()-Assigns a unique sequential number to each customer.
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4. Regional Analysis

Sales by Location

Sales performance was evaluated across:

- Country
- Region
- Market

Sales Contribution

- Contribution of each region to total sales was measured.
- Calculated using SUM(Sales) OVER().
- Helped identify high-performing geographical areas.

5. Shipping Analysis

Delivery Performance Measurement

Shipping efficiency was analyzed using:

- Delivery Time
- Fastest Ship Mode
- Delayed Orders

Delivery Calculation

- Delivery duration was calculated using DATEDIFF().
- Delayed shipments were identified for operational improvement.

6. Profitability Analysis

Discount Impact on Profit

- The relationship between discounts and profitability was evaluated.
- Average profit was measured using AVG(Profit).
- Results were grouped based on discount categories using GROUP BY Discount.
- Analysis helped determine whether higher discounts reduced profitability.

KPI DEVELOPMENT:

- Total Sales → SUM(Sales)
Measures total revenue generated by the business.
- Total Profit → SUM(Profit)
Shows overall business profitability.
- Total Orders → COUNT(Order_ID)
Tracks total customer transactions.
- Total Customers → COUNT(DISTINCT Customer_ID)
Measures customer reach and growth.
- Average Discount → AVG(Discount)
Evaluates effectiveness of discount strategy.

Business KPIs

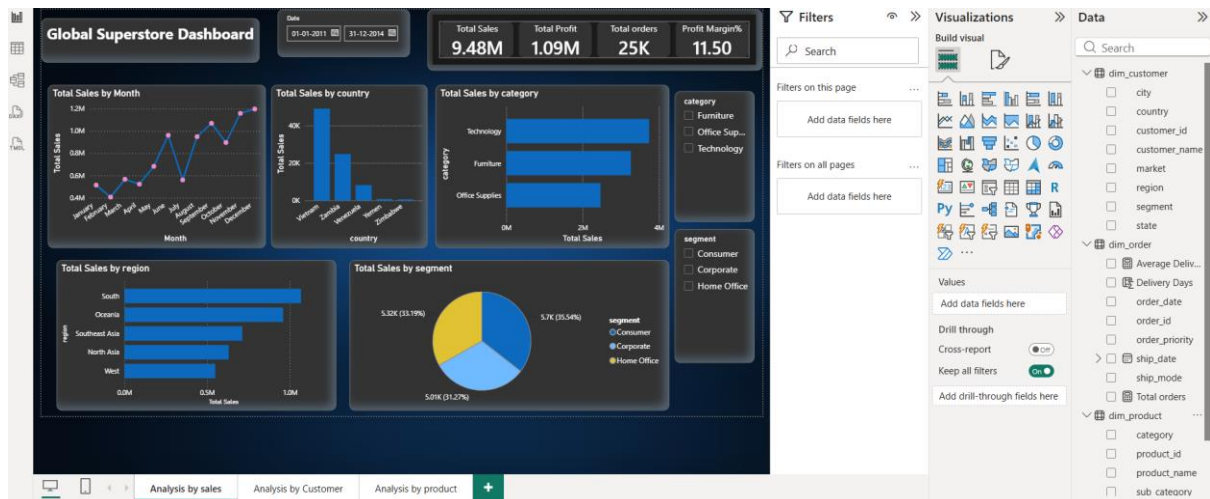
- Revenue KPI → Monitor overall sales performance.
- Profit KPI → Measure business financial health.
- Customer Count KPI → Analyze customer acquisition.
- Order Count KPI → Track business demand.
- Discount Rate KPI → Optimize pricing and profitability.

POWER BI DASHBOARD:

Developed Dashboards:

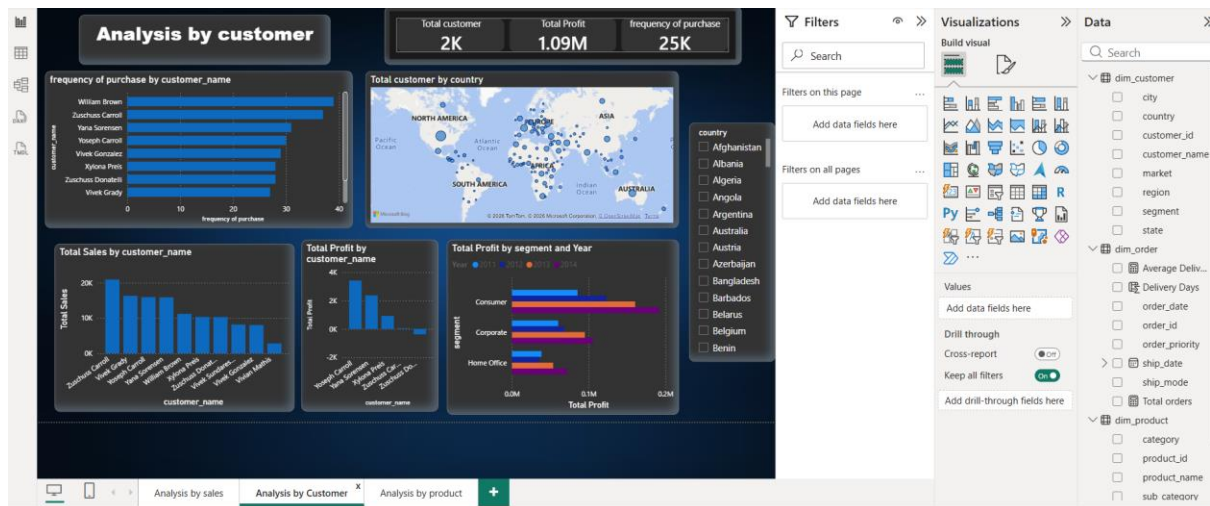
Dashboard 1:

Sales Performance



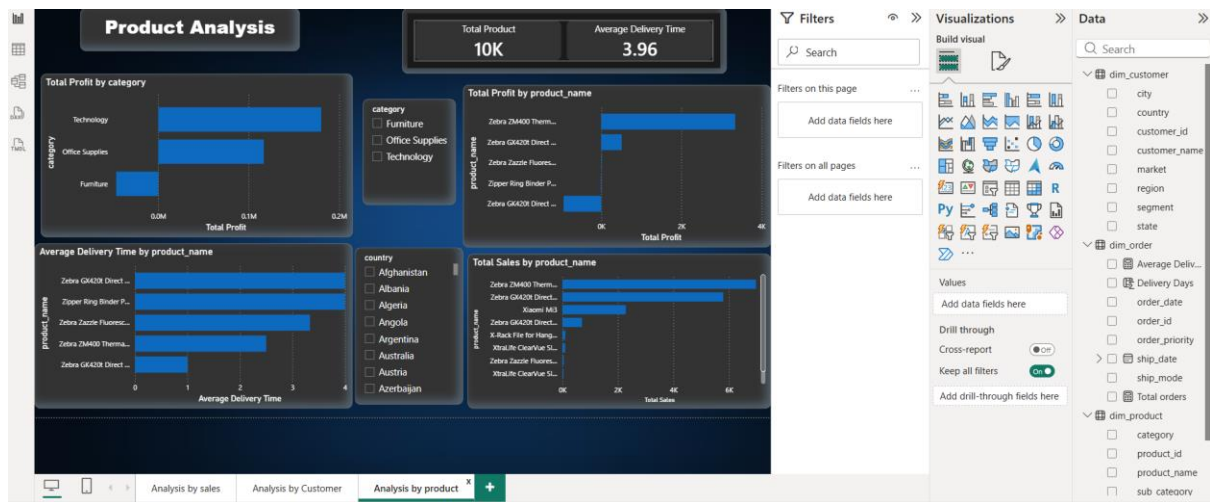
Dashboard 2:

Customer Analysis



Dashboard 3:

Product Analysis:



Key Insights:

- Total revenue exceeded 9 million.
- Profit exceeded 1 million with 11.5% margin.
- Technology is the strongest category.
- Consumer segment contributes the highest business value.
- Certain products and customers generate significantly higher profit than others.
- Sales performance varies considerably by country and region.

Recommendations:

- Increase focus on Technology products.
- Retain high-frequency customers through loyalty programs.
- Improve performance in low-sales regions.
- Review loss-making products and discount strategies.
- Optimize inventory for top-selling products.
- Improve delivery efficiency to enhance customer satisfaction.

Business Impact:

- Helps management identify profitable customers and products.
- Supports data-driven marketing and sales strategies.
- Improves profitability through product and customer segmentation.
- Enables regional performance tracking.
- Helps reduce losses from low-performing products.
- Supports strategic planning and resource allocation.
- Creates measurable business value through actionable insights.

Conclusion:

The Global Superstore project helped identify important insights related to sales, profit, customers, and products. The analysis showed that Technology is the best-performing category and high-frequency customers contribute significantly to revenue. Using SQL and Power BI, business performance was analyzed effectively, helping support better decision-making and profitability. Overall, this project shows how data can be converted into business insights to support better decision-making.